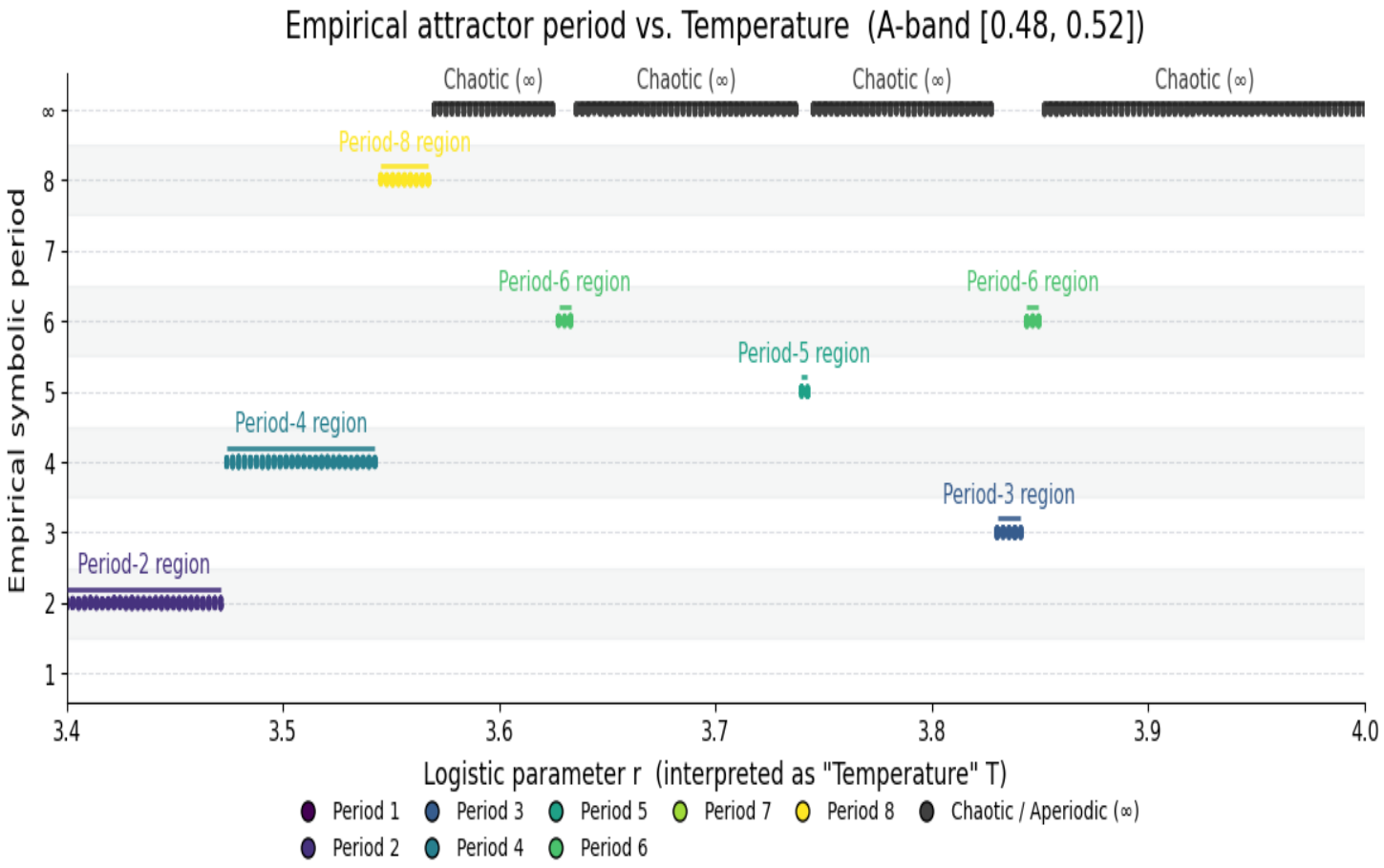
Empirical Attractor Periods vs. Control Parameter in the Logistic Map

- Symbolic dynamics analysis illustrating the period-doubling cascade and periodic windows embedded within the chaotic regime.



How the Logistic Map Reproduces the Empirical Sequence Bands

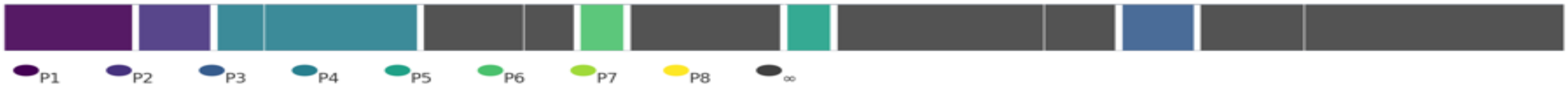
Symbolic A/B/D encoding + minimal-period detection across r ('Temperature')

Setup to match Slide 1

- Encode x_t to A/B/D with an A-band around 0.5: [τ_L=0.48, τ_U=0.52].
- Sweep logistic parameter r ∈ [3.40, 4.00] with high resolution (r_steps=220).
- For each r, run multiple initial conditions (n_inits=30) to reveal fractions of runs.
- Discard transient (≈900) and analyze a tail window (≈240) to settle on the attractor.

Key findings

- We recover the classic period-doubling cascade: 2 → 4 → 8 → ∞.
- Small periodic windows (3, 5, 6, ...) reappear inside the chaotic regime.
- The 'fraction of runs' look emerges naturally from multiple initial conditions.
- Band placement matters: widen [τ_L, τ_U] and A merges; narrow too much and A is missed.



- Increasing the logistic parameter *r* (interpreted as “temperature”) drives the system through a period-doubling route to chaos.
- The empirical A-band sequences correspond closely to these symbolic periods, validating that the logistic map can qualitatively reproduce the structure of the observed attractor transitions.